

IMANE YACOUBI

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

Sep. 2022 – Jul. 2027 

· Mentor, Speaker – *Women in CS* | Advisor, *CS Club* | Orientation Leader, *Math Faculty*

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Go, Cypher

Technologies: AWS, Azure, Node.js, Confluent, Neptune DB, Postman, gRPC, Express.js, Svelte, React

EXPERIENCE

NVIDIA

Sept. 2026 – Dec. 2026

Incoming Software Engineering Intern — Ground Truth signals

Santa Clara, CA

Confluent

May 2026 – Aug. 2026

Software Engineering Intern — Stream Governance

Toronto, ON

- Architected Confluent Cloud's **Stream Lineage V2** in Go, replacing a frontend-only design — which built the lineage graph on every page load and crashed past 1.5k entities — with a time-aware backend service on Amazon Neptune, **eliminating crashes** and introducing point-in-time lineage lookup
- Built a real-time, fault-tolerant ingestion pipeline in Go consuming a gRPC streaming watch API on the Global Resource Server, syncing lineage data within 2 seconds and replacing a 10-minute polling design
- Designed a REST API serving lineage subgraphs at organization, environment, cluster, and resource scope, exposing lineage as a standalone service and **cutting latency by 89%** for the Stream Lineage UI

Microsoft

Jan. 2026 – May. 2026

Software Engineering Intern — Network Tokenization

Redmond, WA

- Designed and built Microsoft's **UCP Payment Handler** that allows Microsoft products to be sold on agent platforms, projected to drive **\$300M+ in revenue** for Microsoft by 2027
- Implemented a PCI-compliant tokenization and credential-binding scheme that conceals card data from agent platforms, **cutting replay-attack risk by 82%**
- Eliminated manual per-agent onboarding by implementing Microsoft Payments' UCP discovery endpoint, enabling agents to dynamically discover checkout capabilities at runtime

Microsoft

May 2025 – Jul. 2025

Software Engineering Intern — Payment Instruments Management

Redmond, WA

- Drove development of a .NET **incident investigation tool** for Microsoft Payments, **reducing engineer triage time by 33%** and enhancing reliability of 50M+ daily payment transactions
- Built a RAG pipeline over a vector DB of historical incident logs, grounding Azure OpenAI models on retrieved findings to surface fixes and uncover patterns for new incidents
- Eliminated manual monitoring of **100+** prod error types across **200K+** daily logs by designing an LLM-powered tool to flag critical issues via trend analysis and auto-create Azure DevOps tasks for them

Maxon Computer

May 2024 – Aug. 2024

Software Engineering Intern — Central Maxon App

Waterloo, ON

- Improved license management for **1.7M users** by integrating 10Duke licensing APIs into Maxon App
- Added service-version tracking to analytics schema/APIs, cutting analytics framework tickets by **40%**

Huawei Technologies

May 2023 – Aug. 2023

Software Engineering Intern — Web Engine Research

Markham, ON

- Owned development of end-to-end benchmark tool in Node.js for WebView performance on mobile, benefiting **100+** developers in Web Engine R&D

RESEARCH

Distributed Systems Lab | Serverless Computing Storage (Advised by Prof. Bernard Wong )

Contributed to a fast state storage system for Serverless functions